Summary

Lego Designers

Subject	Year	Start date	Duration
Design	Grade 6	Week 3, August	6 weeks 40 hours

Description

Students will engage with the design cycle and learn about product design to plan and create a new Lego set. They will develop research, planning, and evaluation skills to help them create a quality design. We will use a 3D design program to construct sets, render images, and create instruction booklets. The skills students learn will lay a foundation for success across MYP and even DP design.

Key and Related Concepts

Key Concepts

Key Concepts	Definition
Systems	**Systems** are sets of interacting or interdependent components. Systems provide structure and order in human, natural and built environments. Systems can be static or dynamic, simple or complex.

Related Concept(s)

Evaluation, Form

Inquiry

Global Context

Global Context	Explorations to develop
Personal and cultural expression	

Statement of Inquiry

Designers must develop research, planning, and evaluation skills to create good products.

Inquiry Questions

Type	Inquiry Questions
Factual	What are some of the main Lego product lines available, and how are they different?
Factual	How does Bricklink Studio help in creating 3D models of Lego sets?
Factual	What are the key elements of product design in the context of Lego sets?
Conceptual	Why is research important before designing a product?
Conceptual	What makes a Lego set engaging to build and play with?
Conceptual	How can user feedback and evaluation improve the design of a Lego set?
Debatable	Why is package design important?
Debatable	Is Lego educational to build with or just a toy? Why or why not?

ATL Skills

ATL skills

Research

- VI. Information literacy skills

Finding, interpreting, judging and creating information

Access information to be informed and inform others

Make connections between various sources of information

Understand the benefits and limitations of personal sensory learning preferences when accessing, processing and recalling information

Use memory techniques to develop long-term memory

Present information in a variety of formats and platforms

Evaluate and select information sources and digital tools based on their appropriateness to specific tasks

Understand and use technology systems

Thinking

- IX. Creative thinking skills

Generating novel ideas and considering new perspectives

Use brainstorming and visual diagrams to generate new ideas and inquiries

Design improvements to existing machines, media and technologies

Apply existing knowledge to generate new ideas, products or processes

Create original works and ideas; use existing works and ideas in new ways

Assessment

Formative Assessment

JUN 11
(FA1) Final box design
Formative 15/17 Students Thursday at 3:25 PM

Upload your FINAL box design here as a PDF.

JUN 11
(FA1) Final box design
Formative 15/17 Students Thursday at 10:15 AM

Upload your FINAL box design here as a PDF.

FEB 2
(FA1) IDU Math Video
Formative Project 9/16 Students Monday at 9:35 AM

Upload your math explainer video by **downloading it** from Canva as an **MP4 video.**

FEB 2
(FA1) IDU Math Video
Formative Project 13/17 Students Monday at 9:35 AM

Upload your math explainer video by **downloading it** from Canva as an **MP4 video.**

SEP 10
(FA1) Design Studio Quiz
Formative Quiz 0/16 Students Wednesday at 1:50 PM

A quiz about Bricklink Studio.

SEP 3
(FA1) Design Studio Quiz
Formative Quiz 0/17 Students Wednesday at 9:50 AM

A quiz about Bricklink Studio.

Summative Assessment

JUN 1
(SA1) Design Evaluating your Work Criteria D
Summative Report 13/17 Students Monday at 3:30 PM

Take a picture of your work and upload to Managebac, turn in your paper in class.

MAY 29
(SA1) Design Evaluating your Work Criteria D
Summative Report 0/17 Students Friday at 3:30 PM

Take a picture of your work and upload to Managebac, turn in your paper in class.

MAY 14
(SA1) Design Criteria C Creating My Lego Set
Summative Project 15/17 Students Thursday at 10:10 AM

Upload your **PDF** from Canva, your **BOX DESIGN PDF,** your **2 PAGES OF INSTRUCTIONS,** AND your **STUDIO FILE (.io).**

MAY
(SA1) Design Criteria C Creating My Lego Set

13
Summative Project 17/17 Students Wednesday at 3:35 PM

Upload your **PDF** from Canva, your **BOX DESIGN PDF,** your **2 PAGES OF INSTRUCTIONS,** AND your **STUDIO FILE (.io).**

APR 13
(SA1) Design Lego Set Ideas
Summative Project 14/17 Students Monday at 3:30 PM

Students complete their assessment work in Canva and upload a PDF to ManageBac once finished.

APR 7
(SA1) Design Lego Set Ideas
Summative Project 15/17 Students Tuesday at 3:30 PM

Students complete their assessment work in Canva and upload a PDF to ManageBac once finished.

MAR 13
(SA1) Design Lego Research
Summative Project 15/17 Students Friday at 10:15 AM

Students will complete and upload their Lego research task sheet.

MAR 12
(SA1) Design Lego Research
Summative Project 0/17 Students Thursday at 10:15 AM

Students will complete and upload their Lego research task sheet.

DEC 19
(SA1) Design Evaluating your Work
Summative Report 12/16 Students Friday at 9:20 AM

Complete your Lego Evaluation SA worksheet. Take a picture of each side and upload to ManageBac.

DEC 12
(SA1) Design Evaluating your Work
Summative 15/17 Students Friday at 9:20 AM

Complete your Lego Evaluation SA worksheet. Take a picture of each side and upload to ManageBac.

NOV 26
(SA2) Design Creating Your Set
Summative Project 15/17 Students Wednesday at 12:20 PM

Complete your Lego project and submit all materials to ManageBac!

NOV 14
(SA2) Design Creating Your Set
Summative Project 15/16 Students Friday at 12:25 PM

Complete your Lego project and submit all materials to ManageBac!

OCT 16
(SA1) Design Planning your Lego Set
Summative Project 14/16 Students Thursday at 3:30 PM

Students complete the assigned Canva presentation to plan and prepare to create their new Lego set. Upload as a **PDF.**

OCT 16
(SA1) Design Planning your Lego Set
Summative Project 15/17 Students Thursday at 10:20 AM

Students complete the assigned Canva presentation to plan and prepare to create their new Lego set. Upload as a **PDF.**

SEP 22
(SA2) Design Research A
Summative Paper 0/16 Students Monday at 11:10 AM

Students will hand in their completed criteria A pamphlets.

SEP 22

(SA2) Design Research A

Summative Paper 0/17 Students Monday at 11:10 AM

Students will hand in their completed criteria A pamphlets.

Learning Experiences

Learning Experiences and Teaching Strategies

Day	Lesson Focus
1	Introductions; writing sample for academic integrity SWBAT: complete a 'get to know me' writing page How: fill out a form provided in class Why: Academic integrity reference Links: N/A
2	Introduction to Bricklink studio, Moodle SWBAT: Complete tutorial in Studio, open Moodle for exit ticket How: Follow tutorial steps in Studio, provide easy instructions for extension Why: Important to learn where to find resources and tasks Links: N/A
3	Bricklink Studio building with instructions SWBAT: Assemble an official Lego set in Studio How: Download provided instructions to build along Why: Helps to learn typical Lego construction as well as where to find pieces Links: Instructions,
4	Bricklink Studio guided lesson 2: Building a six sided die SWBAT: Flip pieces around and manipulate them with shortcuts How: They follow the teacher for each step Why: Core Studio and 3D skills Links:Instructions,

5	Bricklink Studio building projects SWBAT: follow instructions to build sets How: Download provided instructions to build along Why: Helps to learn typical Lego construction as well as where to find pieces Links: Instructions,
6	Bricklink Studio building projects: add rendering and special colors SWBAT: follow instructions to build sets How: provide instructions to students Why: introduce rendering Links: Instructions, Studio references
7	Bricklink Studio guided lesson 5: Part Designer (Custom pieces) SWBAT: build custom pieces in part designer How: use part designer and follow teacher example Why: Discover how to add totally custom pieces to their work Links: Instructions, Studio references
8	Introduce Criteria A: Formative Research Task SWBAT: identify elements on Lego website How: complete a provided example research task Why: Learn skills to be used on SA Links: N/A
9	Criteria A: Summative Introduction SWBAT: Understand what tasks will be completed during assessment How: work through SA with students Why: Prepare them to succeed at SA Links: Criteria A assessment
10	Criteria A: Research Tasks Day 1 SWBAT: choose a Lego set to explore How: Use Lego website or Bricklink Studio to find a set

	Why: Learn what to do in assessment Links: Brickipedia, Lego.com, Criteria A assessment
11	Criteria A: Research Tasks Day 2 SWBAT: Complete analysis for 4 different sets How: Find sets that will help inspire your work Why: Determine project direction Links: Brickipedia, Lego.com, Criteria A assessment
12	Criteria A due at end of class SWBAT: Complete their paper assessment for research How: FInish their research tasks Why: Prepares them to move into planning Links: Brickipedia, Lego.com, Criteria A assessment
13	Design Challenge: Build a set from images SWBAT: create as much of a set as possible from images How: provide image references but no instructions Why: Useful to gauge piece recognition Links: Brickipedia, Lego.com,
14	Introducing Criteria B: Formative Planning Task SWBAT: engage with ideas from the assessment How: create specifications and success criteria for a sample project Why: prepares students for criteria B assessment Links: N/A
15	Criteria B: Summative Introduction SWBAT: begin working on their planning assessment How: Use Canva assessment Why: Develop individual specifications for project Links: Assessment B in OneDrive
16	Criteria B: Planning Tasks Day 1

	SWBAT: complete success criteria How: refer to sample criteria and prepare their own Why: establish individual criteria for their set Links: Assessment B in OneDrive
17	Criteria B: Planning Tasks Day 2 SWBAT: complete inspiration page How: Use images from Lego or Brickipedia Why: Help identify pieces and construction methods Links: Assessment B in OneDrive
18	Criteria B: Planning Tasks Day 3 SWBAT: Work on rough draft How: Use studio to build out ideas Why: Hands on with methods and pieces Links: Assessment B in OneDrive
19	Criteria B: Planning Tasks Day 4 SWBAT: continue rough draft of parts of their set How: Use studio to build out ideas Why: Hands on with methods and pieces Links: Assessment B in OneDrive
20	Criteria B due at end of class SWBAT: complete design specifications How: Use canva to finish the work, rough draft in Studio Why: Ensures all elements are in place for C Links: Assessment B in OneDrive
22	Introduce Criteria C: Formative Rendering Tasks SWBAT: complete renders that attempt to match specific images How: provide image references and test their recall of rendering Why: necessary to produce attractive box designs

	Links: N/A - produce in studio or use a real Lego set
23	Criteria C: Summative Creating Tasks Day 1 SWBAT: Start building their Lego set in Studio How: Documenting changes, sharing images, and piecing together a set Why: Teaches 'showing their work', and proceeding thoughtfully Links: Canva link
24	Criteria C: Creating Tasks Day 2 SWBAT: Continue building Lego set in Studio How: Documenting changes, sharing images, and piecing together a set Why: Slowing down and learning the build process Links: Canva link
25	Criteria C: Creating Tasks Day 3 SWBAT: Continue building their set How: Documenting changes, sharing images, and piecing together a set Why: Continued build process Links: Canva link
26	Criteria C: Creating Tasks Day 4 SWBAT: Continue building their set How: Documenting changes, sharing images, and piecing together a set Why: Continued build process Links: Canva link
27	Criteria C: Creating Tasks Day 5 SWBAT: Continue building their set How: Documenting changes, sharing images, and piecing together a set Why: Continued build process Links: Canva link
28	Criteria C: Creating Tasks Day 6 SWBAT: Continue building their set

	How: Documenting changes, sharing images, and piecing together a set Why: Continued build process Links: Canva link, Box design template
29	Criteria C: Creating Tasks Day 7 SWBAT: Continue building their set How: Documenting changes, sharing images, and piecing together a set Why: Continued build process Links: Canva link, Box design template
30	Criteria C: Creating Tasks Day 8 SWBAT: Begin rendering work for their box design How: Documenting changes, sharing images, and piecing together a set Why: Moving rough project to finished state Links: Canva link, Box design template
31	Criteria C: Creating Tasks Day 9 SWBAT: complete renders for box design, create instructions How: use render tool in studio Why: Moving project to completed state Links: Canva link, Box design template
32	Criteria C due at end of class SWBAT: finish all parts of criteria C assessment How: Complete work in studio Why: Ensure taks for all criteria are complete Links: Canva link, Box design template
33	Criteria C polishing up projects 1 SWBAT: revise work before presenting to others How: Studio, box design template Why: Ensure work is quality for others Links: Canva link, Box design template

34	Criteria C: Polishing projects day 2 SWBAT: revise work before presenting to others How: Studio, box design template Why: Ensure work is quality for others Links: Canva link, Box design template
35	Criteria D: Summative Evaluating Day 1 SWBAT: Choose testing methods for their work How: Use created interview question site Why: qualitative questions will help students gather useful data Links: Question helper, Criteria D assessment
36	Criteria D: Evaluating Summative Day 2 SWBAT: Interview people at school to collect data How: Walk around in groups and present work to others Why: Important to learn how to interview and collect feedback Links: Criteria D assessment
37	Criteria D: Evaluating Summative Day 3 SWBAT: assess their work based on success criteria How: Review their personal success criteria and respond to them Why: They need to answer whether they followed their own guidelines Links: Criteria D assessment
38	Criteria D due at end of class SWBAT: identify improvements they can make and critically evaluate the set How: Finish part 3 and 4 of the assessment Why: Part of design cycle Links: Criteria D assessment
39	Final Revisions Day 1 SWBAT: students incorporate feedback and evaluated info into designs How: Revisions based on Criteria D Why: Completing the design cycle

	Links: Criteria D assessment
40	Final Revisions Day 2 SWBAT: students incorporate feedback and evaluated info into designs How: Revisions based on Criteria D Why: Completing the design cycle Links: Criteria D assessment
41	Final Revisions Day 3 – upload at end of class SWBAT: students incorporate feedback and evaluated info into designs How: Revisions based on Criteria D Why: Completing the design cycle Links: Criteria D assessment
42	Design Challenges

Reflections

General Reflections

Prior to studying the unit

Paul Kwiatkowski Sep 1, 2025 at 8:47 AM

This unit went pretty well last year, and was a very strong evolution of the Lego Architects unit. This year I will be incorporating a bit more 'serious' content by way of the smallest bit of DP connections. I also plan to devote more time to 'speed challenges' to shake up students and reduce hyper-fixation on the single end product. One challenge from before was keeping students on schedule. They were usually on task, but the task ran long. This time I will build in additional assessment deadlines to ensure students complete one part then move on, instead of dragging things out. I also plan to break apart the assessments a bit, both for ease of grading, and to make very clear what criteria is being assessed.

During the unit

Paul Kwiatkowski Nov 3, 2025 at 9:27 AM

Revisions have been successful so far in helping to guide student work. Students did struggle a bit with identifying a need for their product, but most were able to understand the idea upon further explanation. Broadly, students have completed all the tasks in the assessments, guided by the 'fill in this page/blank' type of organization.

After the unit

Paul Kwiatkowski Jan 19, 2026 at 8:24 AM

We identified a few revisions to make to task criteria to make grading more clear. I realized that there wasn't a clear explanation of technical skill, and while that is often nebulous, on this project it can be much more clearly laid out. I need to deliver a completed example packet to students at the start of each assessment to help demonstrate expectations. Students did enjoy the unit considerably, and are looking forward to challenges to further demonstrate their skill.

Paul Kwiatkowski Jun 15, 2026 at 11:20 AM

I leaned more heavily on Lego instructions for training this semester and generally found them to be a good tool. They can present challenges and a larger volume of questions (piece is missing!), but far less students were 'behind' compared to build-a-long projects.